DOCKHUB · USER REQUIREMENTS

DockHub user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First user requirements, written at the kit 1.9.1 upgrade from the users and stakeholders in DKH-PRB-001 (Tables 1 and 2); they score the concept options and every product requirement in DKH-PRS-001 traces to one

DockHub user requirements

What the people who use, host, service and regulate a DockHub station need, written from their side. They come from Tables 1 and 2 of the problem statement (DKH-PRB-001) and are desk research until the co-design sessions the problem statement lists have been run with a partner (Los Deliveristas Unidos is the first candidate, DKH-DEC-001). They score the concept options (DKH-OPT-001), and each product requirement in DKH-PRS-001 traces to one of them (DKH-RTM-001). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Usability	Fast swap	I put in my flat battery and take a charged one in about a minute, in a hurry, with gloves, at night	High	App-based delivery rider (PRB Table 1)
UR-002	Performance	A charged battery is there	When I arrive in a busy hour there is a charged battery, for 20 or more riders a day	High	Delivery rider; courier fleet operator
UR-003	Safety	No fire danger	The station does not set fire to the street, the building or the riders, and if a battery fails the harm stays inside it	High	City fire safety department; host; rider
UR-004	Environment and durability	Works outdoors all year	It works in rain, dust, sun, road salt and from -10 to 45 C	High	Host; transport department
UR-005	Interoperability	Uses the open battery	It takes SwapCell packs as they are, with no change to the pack	High	Fleet operator; SwapCell project
UR-006	Safety	Stays up in a storm	It stays upright in a 30 m/s gust with its canopy	High	Transport department; host
UR-007	Security	Hard to rob or tamper with	A battery is released only to a valid user who returned one; the cabinet resists casual attack	Medium	Host; fleet operator
UR-008	Usability	Everyone can use it	Controls are reachable and readable by people of all heights and abilities and in the dark	Medium	Transport department; rider
UR-009	Logistics	Small and easy to power	It takes little sidewalk and one ordinary circuit	Medium	Host; transport department
UR-010	Privacy	No watching me	No camera or microphone; the station knows a battery, not who I am; it keeps working offline	Medium	Rider; city
UR-011	Maintenance	Quick to mend	One technician mends any part with hand tools in a short visit	Medium	Technician; fleet operator
UR-012	Cost	Affordable to build	A cooperative, fleet or small business can afford to build one	Medium	Fleet operator; cooperative
UR-013	Environment and durability	Some of it from the sun	Part of the charging energy comes from the canopy	Low	Transport department

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory, interoperability, security, privacy. Priorities: High, Medium, Low.